A LEAN FORMALIZATION OF THE LONELY RUNNER CONJECTURE FOR EIGHT RUNNERS

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Abstract. For positive integer speeds $v_1, \dots, v_n$ let $\mathrm{ML}(v) = \sup_{t\in\mathbb{R}} \min_i \|v_i t\|$, where $\|x\|$ is the distance from x to the nearest integer. The lonely runner conjecture for $n+1$ runners asserts $\mathrm{ML}(v) \geq 1/(n+1)$. We describe a formalization in Lean 4 with Mathlib of the case $n = 7$ (eight runners), proved by Rosenfeld in 2025. A module outside any lake build proves the statement from the three standard axioms of Lean alone: it imports 27 kernel proofs of finite certificates, kept as object files and checked by the Lean kernel. The theorem in the module `EightRunners` states the same result but depends in addition on 27 axioms, each recording one of these kernel checks after the object files had been removed for disk reasons. The certificates are regenerated by a published procedure and compared with recorded SHA-256 hashes. On the way the formalization proves the case $n = 6$ (seven runners, Barajas and Serra, 2008) from the standard axioms alone, and replaces the height bound of Malikiosis, Santos and Schymura used by Rosenfeld with a pigeonhole argument and the planar rounding step of Giri and Kravitz. The formalization of Barajas and Serra found errors in eleven places of the printed proof; none affects the theorem. No new mathematical result is claimed. No whole-lake rebuild was run for this release, and the Lean sources are provided as is.

1. Statement and trust base

1.1. **Statement.** For $v \in \mathbb{Z}^n_{>0}$ and $t \in \mathbb{R}$ put $\mathrm{gap}_v(t) = \min_i \|v_i t\|$ and $\mathrm{ML}(v) = \sup_t \mathrm{gap}_v(t)$. The lonely runner conjecture of Wills [17] and Cusick [5] asserts

$$\mathrm{LRC}(n): \qquad \mathrm{ML}(v) \geq \frac{1}{n+1} \quad \text{for all } v \in \mathbb{Z}^n_{>0}.$$

It is known for $n \leq 6$ [3, 2], for $n = 7$ [11], for $n = 8, 9$ [12, 16], and beyond [13, 1]. In Lean, speeds are v : `Fin k` $\to \mathbb{Z}$, $\|x\|$ is `nearestIntDist` $x = \min(\{x\}, 1 - \{x\})$, and ML is the conditional supremum `sSup (Set.range (gap v))`. Repeated speeds are allowed; no primitivity is assumed.

Theorem 1.1 (Lean `LRSpectrumFull.EightRunners.eight_runners_unconditional`)**.** *For every v : `Fin 7` $\to \mathbb{Z}$ with $v_i > 0$ for all i,* $\mathrm{ML}(v) \geq 1/8$.

Theorem 1.2 (Lean `LRSpectrumFull.BarajasSerra.LRC6_barajasSerra`)**.** *For every u : `Fin 6` $\to \mathbb{Z}$ with $u_i > 0$ for all i,* $\mathrm{ML}(u) \geq 1/7$.

The module `EightRunners.lean` contains only Theorem 1.1. It imports `EightE5`, which imports the reduction, the certificate axioms and Theorem 1.2.

1.2. **Trust base.** Table 1 lists the output of `#print axioms`, run on 2026-09-27 with `lake env lean` after building the module `EightRunners`. The literal output is reproduced in the file `LEAN_MAP.md` accompanying this paper.

The 27 axioms state `lemma7Check 7 p = true` for the primes $p \in S = \{31, 37, 43, \dots, 157, 163\}$ of Rosenfeld's set, $|S| = 27$. For each p the statement was proved by the Lean kernel (`decide +kernel`) from a generated certificate, and `#print axioms` of that proof reported the three standard axioms only. The generated sources (about 1.6 GB in total) and their object files

Declaration	Axioms beyond `propext`, `Classical.choice`, `Quot.sound`
`EightRunners.eight_runners_unconditional` (Thm. 1.1)	the 27 axioms `EightCert27.lemma7Check_seven_p`*p*`_verified`
`EightCert27.hchk_all` (all 27 primes)	the same 27
`EightRunnersReal.eight_runners_unconditional_real` (same statement, outside any lake build)	none
`EightRunnersReal.hchk_all_real` (all 27 primes)	none
`BarajasSerra.LRC6_barajasSerra` (Thm. 1.2)	none
`EightE5.mssGap7_holds` (rounding step)	none
`EightMSS.e5b_staircase` (pigeonhole)	none
`EightV4.lemma7Check_of_checkRootH` (checker soundness)	none

TABLE 1. Trust base. No declaration listed depends on `sorryAx`; no `native_decide` (`Lean.ofReduceBool`) is used.

were then deleted, and the fact was reinstated as a named axiom whose docstring records the SHA-256 hash of the concatenated sources. These axioms record kernel checks run by the author; the evidence that each check succeeded is the per-prime record in `code/eight/records`, and the axioms are not re-checked by a build of the present tree. All 27 certificates have been regenerated and re-checked by the kernel at least once (25 of them on 2026-09-27, the remaining two, $p = 31$ and $p = 37$, earlier), with hashes matching the recorded originals; the records in `code/eight/records` give the kernel results. Section 3.2 gives the procedure that regenerates each certificate, compares its hash, and rebuilds its kernel proof.

The module `EightRunnersReal.lean` removes the 27 axioms. It imports the modules `E8Stage.P`*p*`.Asm` produced by procedure (2) of Section 3.2 with the object files kept (about 2.6 GB in total, outside the lake package, regenerable in about 12 hours), restates `hchk_all_real` from the 27 theorems `EightV4.Cert`*p*`.lemma7Check_seven_p`*p*, and proves `eight_runners_unconditional_real` through `EightE5.eight_runners_of`, which takes the per-prime checks as a hypothesis. On 2026-09-30, compiled with `lean` against these object files, `#print axioms` reports exactly `propext`, `Classical.choice`, `Quot.sound` for both declarations. The run was repeated on 2026-11-01 (raw `lean` 4.30.0, one process, 50 to 110 s with a cold file cache, peak memory footprint 1.08 GB, maximum resident set 6.4 to 7.1 GB, the latter being mostly memory-mapped object files); the output was the same for both declarations, for the 27 theorems `EightV4.Cert`*p*`.lemma7Check_seven_p`*p* and for the intermediate theorems `EightE5.eight_runners_of`, `EightE5.mssGap7_holds`, `BarajasSerra.LRC6_barajasSerra`, `EightPlan.e2_prime_dvd`, `EightPlan.e3_lcm_dvd`, `EightPlan.e4_parity`. Three declarations of the import closure whose bodies are `sorry` (`BarajasSerra.primeFilteringCor3`, `EightMSS.bridge31`, `EightPlan.e5_sum_STUB_SUPERSEDED`) print `sorryAx` in the same run; none of them is used by the theorem. The generator is not part of the trust base: a certificate that the checker rejects cannot yield a theorem, and the recorded hashes serve reproducibility only. What is trusted is the Lean kernel, Mathlib at the pinned revision, and the statement of `ML`. As a control, the same environment reports the 27 `EightCert27` axioms for `EightRunners.eight_runners_unconditional`. The module `EightE5` imports `EightCert27`, so the 27 axioms

are present in that environment but are not used by the new theorem. This module is not built by `lake build`; it is compiled by a script in `code/eight/stage/`. For the primes $p \in \{31, 37, 43, 47, 53, 59, 61, 67, 113, 163\}$ the stored declarations were in addition replayed through a fresh kernel with `Lean.Environment.replay`, from the kept object files, against the environment of their imports; all were accepted with the three standard axioms and the statement `lemma7Check 7 p = true`. The remaining 17 primes were not replayed. This shows that the stored terms re-typecheck; it does not audit Lean, Mathlib or the checker `lemma7Check`.

2. Rosenfeld's reduction as formalized

We follow [11, Section 5] with $n = 7$. Lean names in this section are in the namespace `LRSpectrumFull.EightPlan` unless stated otherwise. A counterexample is a primitive $v \in \mathbb{Z}^7_{>0}$ with $\mathrm{ML}(v) < 1/8$ (`IsCE`).

- E0 (gcd reduction) ML is invariant under division by $\gcd(v)$, so it suffices to treat primitive v (`e0_primitive`).
- E4 (leave-one-out, [11, Lemma 5]) Given LRC(6), every six of the seven speeds of a counterexample are coprime; in particular at most five of them are even (`e4_leaveOneOut`, `e4_parity`).
- E2 ([11, Lemmas 6, 7]) If `lemma7Check 7 p = true` then p divides some speed of every counterexample satisfying the parity condition of E4 (`e2_prime_dvd`). Here `lemma7Check` is a Boolean function defined by enumeration of admissible residue tuples modulo p.
- E3 ([11, Lemma 4]) Consequently $840 \prod_{p \in S} p$ divides $\prod_i v_i$ (`e3_lcm_dvd`).
- NUM $28^{42} < 7^7 \cdot 840 \prod_{p \in S} p$ (`numeric`), and by the inequality of arithmetic and geometric means $28^{42} < 7^7 \prod_i v_i$ implies $\sum_i v_i > 28^6$ (`e5_amgm`).
- E5 Given LRC(6), a primitive $v \in \mathbb{Z}^7_{>0}$ with $\sum_i v_i > 28^6$ has $\mathrm{ML}(v) \geq 1/8$ (Section 4).
- E6 LRC(6): Theorem 1.2 (Section 5).

The assembly is `EightE5.eight_runners_of`, which takes LRC(6) and the 27 prime checks as hypotheses and uses only the standard axioms; `EightE5.eight_runners_unconditional` instantiates it with Theorem 1.2 and `EightCert27.hchk_all`. A counterexample then satisfies E3, hence $\sum v_i > 28^6$ by NUM, contradicting E5.

3. The prime campaign

3.1. **The certificate checker.** Evaluating `lemma7Check 7 p` directly in the kernel is infeasible for the larger primes. We wrote a second checker, `EightCheckV4`, which walks a search tree supplied as data (a certificate) and accepts only if every leaf is closed by a counting bound or a table lookup. Its soundness theorem `EightV4.lemma7Check_of_checkRootH` (file `EightCheckSound.lean`) states that acceptance implies `lemma7Check 7 p = true`; it depends on the standard axioms only. A Rust program generates, for each prime, the certificate as Lean source split into modules; the kernel evaluates the checker on it by `decide +kernel`. The soundness proof does not depend on the generator: a wrong certificate is rejected and cannot produce a false theorem.

3.2. **Reproduction.** For each $p \in S$ a record file stores the generator command, the SHA-256 of the generator, of the checker and of the soundness file, of each generated module and of their concatenation, the toolchain (`leanprover/lean4:v4.30.0`), and the axiom output of the kernel proof. The generator, the scripts and the 27 records are in `code/eight/`. Two procedures are provided.

(1) Hash check (`stage/repro_check.sh p`): regenerate the sources and compare the concatenated hash with the record and with the axiom docstring. This takes seconds per prime and does not run Lean.
(2) Full re-check (`stage/run_prime.sh p`): regenerate, compile every module of that prime with one `lean` process per module, and require that `#print axioms` of `EightV4.Cert`p`.lemma7Check_seven_p`p reports exactly the three standard axioms.

On 2026-09-27 procedure (1), run from `code/eight/`, reproduced all 27 hashes, and every hash occurs in `EightCert27.lean`. Procedure (1) was repeated on a copy of `code/eight/` on 2026-10-18 and again reproduced all 27 hashes (1143 s; `data/repro_logs/eight_repro_all_27.txt`). Procedure (2) was not repeated for the archive. Procedure (2) was run once per prime during the campaign (2026-09-24 to 2026-09-26). Three records ($p = 31, 37, 43$) name an earlier revision of the generator source; the present generator reproduces their hashes.

4. The height bound E5

Rosenfeld obtains E5 from [9, Theorem A′], whose proof passes through zonotopes and successive minima. We did not formalize that proof. E5 is instead obtained from two statements: a pigeonhole lemma and the planar rounding step of Giri and Kravitz [7, Section 7], which reduces a speed vector close to a rational plane to LRC($n-1$). The same step is the case $r = 1$ of [1, Proposition 3.4]. We claim no novelty for either.

Lemma 4.1 (Lean `EightMSS.e5b_staircase`)**.** *Let $v \in \mathbb{Z}^n_{>0}$ be primitive and $l \geq 1$ an integer with $\sum_i v_i > l^{n-1}$. There are $z \in \mathbb{Z}^n$, not proportional to v, and $\sigma \in \mathbb{Q}$ with $|lz_i - \sigma v_i| \leq 1$ for all i.*

Sketch. For each pair $e = (i, a)$ with $0 \leq a < v_i$ put $\tau_e = (2a+1)/(2v_i)$ and let $q_e \in \mathbb{Z}^n$ be the coordinatewise rounding of $\tau_e v$ with ties broken according to the index order, so that the $\sum_i v_i$ points q_e are distinct and $|q_{e,j} - \tau_e v_j| \leq 1/2$. The map $(e, c) \mapsto q_e + cv \bmod l$ on pairs with $0 \leq c < l$ has $l \sum_i v_i > l^n$ arguments, so two arguments collide. If they share e, primitivity gives $l \mid c - c'$, impossible. Otherwise $lz = q_e - q_{e'} - (c' - c)v$ defines $z \in \mathbb{Z}^n$, and $\sigma = \tau_e - \tau_{e'} - (c' - c)$ satisfies the bound. If z were proportional to v, primitivity would give $q_e - q_{e'} = sv$ with $s \in \mathbb{Z}$; the bounds $0 \leq q_{e,k}$ and $q_{e,k} < v_k$ at the index k of e (and of e') force $s = 0$, contradicting $q_e \neq q_{e'}$. □

Proposition 4.2 (Lean `EightE5.mssGap7_holds`)**.** *Assume* LRC(6). *Let $v \in \mathbb{Z}^7_{>0}$, $z \in \mathbb{Z}^7$ not proportional to v, and $\sigma \in \mathbb{Q}$ with $|28z_k - \sigma v_k| \leq 1$ for all k. Then* $\mathrm{ML}(v) \geq 1/8$.

Proof. Choose i, j with $z_i/v_i < z_j/v_j$ adjacent among the distinct values of z_k/v_k (`EightE5.exists_adjacent_pair`). Put $c = (z_i + z_j)v - (v_i + v_j)z \in \mathbb{Z}^7$. Then $c_k = 0$ would mean $z_k/v_k = (z_i + z_j)/(v_i + v_j)$, which lies strictly between the two adjacent values; so no c_k vanishes. Also $c_j = -c_i$. Apply LRC(6) to the six positive integers $|c_k|$, $k \neq j$: there is $\tau \in \mathbb{R}$ with $\|c_k \tau\| \geq 1/7$ for $k \neq j$, and then also for $k = j$. Let $s = \sigma/28$ and $d = z - sv$, so $|d_k| \leq 1/28$. Let N be an integer nearest to $(v_i + v_j)\tau$ and $r = N - (v_i + v_j)\tau$, so $|r| \leq 1/2$. For $t = (z_i + z_j)\tau + rs$,

$$tv_k = c_k\tau + \big((v_i + v_j)\tau + r\big)z_k - rd_k = c_k\tau + Nz_k - rd_k,$$

hence $\|tv_k\| \geq 1/7 - 1/56 = 1/8$ for every k. □

With $l = 28$ and $n = 7$, Lemma 4.1 and Proposition 4.2 give E5 (`EightMSS.e5_sum_of_gap`, `EightE5.e5_sum`, and the product form `EightE5.e5_prod`). The threshold 28^6 is the one Rosenfeld uses, so no other part of the reduction changes.

Remark 4.3. An earlier version of this formalization carried E5 as an axiom transcribing the case $n = 7$ of [9, Theorem A′] (`EightMSSImport.mss_theorem_A_n7`), and an attempt to formalize that theorem through its zonotope argument (`EightZonotope.lean`) stopped at [9, Propositions 3.3, 3.4]. Neither is in the closure of Theorem 1.1; both files are kept and marked as superseded.

5. Seven runners: Barajas and Serra

Lean names are in `LRSpectrumFull.BarajasSerra`. Throughout p is a prime, $N = p^{m+1}$, $\nu = \nu_p$, $q(x) = \lfloor (x \bmod N)/p^m \rfloor \bmod p$ (the arc of length N/p containing x), and Λ_j are the multiplier families of [2, Section 2].

5.1. **Prime filtering.** [2, Lemma 2] (`primeFiltering`) drives the proof: after rescaling so that the speeds sit at p-adic levels $0 \le i \le m$, a counting induction over levels produces a multiplier coprime to p that moves every speed out of the arcs $q \in \{0, p-1\}$, provided each level below the top carries few enough speeds. We formalized the digit arithmetic (q under multiplication by Λ_j, carries of $q(jd)$ for small j, reflection $q(N-d) = p-1-q(d)$), the descending induction, and the passage from $q \notin \{0, p-1\}$ to $\|td\| \ge 1/p$ for $t = \lambda/N$.

5.2. **Three and five runners: finite tables.** For $n = 2$ ($p = 3$) filtering and two short cases suffice (`barajasSerra_card2`). For $n = 4$ ($p = 5$) the configuration with three speeds at a common level and one above is not covered by filtering. We did not follow the paper's case analysis there. Each level-0 speed is replaced by a finite model (its arc, its residue mod 5, and three carry bits), and the claim that any three model elements share a good multiplier $(1+5^m k)j$ with $1 \le j \le 4$, $k < 5$ is checked by the kernel on the model (`BarajasSerraCard4Table`, eight chunks); the theorem is `barajasSerra_card4`. Both cases use the standard axioms only.

5.3. **Seven runners.** For $n = 6$, $p = 7$. After scaling the minimum level to 0, either every level below the top carries at most three speeds, and filtering applies, or one of four distributions occurs: (a) five speeds at level 0 and one above; (b) four at level 0 and two at the top; (c) four at level 0, one strictly between, one at the top; (d) one at level 0, four at an intermediate level, one at the top. The level split and dispatch are `Sec5.lrc6_nat` and `Sec5.lrc6`.

Cases (b)–(d), Section 5 of [2]. The core statement is that four level-0 units admit a common multiplier $(1 + 7^m k)j$, $j \le 5$, moving all four off the arcs 0 and 6. We prove it through a Farey model of the carries of $q(jd)$, $j \le 5$, and small kernel tables (`BarajasSerraSec5Fin`), assembled in `Sec5.sec5`. Cases (b), (c), (d) reduce to it (`Sec5.case_b0`, `case_c0` with the paper's multiplier $u'(1 + 7^{m-i_0})$, `case_d0` with filtering below level i_0).

Case (a), Sections 6 and 7 of [2]. For $m = 1$ the case is decided by a kernel table modulo 98 (`lrc6_m1_scaled`, `BarajasSerraLRC6M1Table`, 17 chunks). For $m \ge 2$ we formalized Lemmas 4, 5, 6, 7, 9, 10, 11, 12 and subsections 6.1 to 6.6 (`Sec6.sec61_witness` to `Sec6.sec66_witness`). Each subsection fixes a pattern of residue classes mod 7 of the five level-0 speeds and concludes that a multiplier coprime to 7 moves all six speeds off the arcs 0 and 6. Statements about $\mathbb{Z}_7$ alone are proved by `decide` (`BarajasSerraZ7`). The reduction of an arbitrary configuration to one of the seven patterns (sign flips, choice of base element) is `Sec6.g4`. Case (a) is `case_a`.

5.4. **Method.** Each lemma of Sections 5 and 6 of [2] was first tested in Rust with exact integer arithmetic on a transcription of the Lean definitions, in most cases exhaustively at $m = 2$ and exhaustively or by sampling at $m = 3$, together with a negative control that must fail. Only then was Lean written. The checkers cited by the errata are in `code/eight/checks/` with the commands that run them.

6. Errata to Barajas and Serra

The file `errata_barajas_serra.md` lists, for each item, the printed statement, the corrected statement, the computational evidence and the Lean declaration. In summary: Lemma 4(ii) has the wrong range in the equal-residue case; Lemma 5 is false as printed for two profiles (not used there); in Section 6 there are wrong intermediate values or signs in 6.1, 6.3, 6.5, 6.6(a) and 6.6(b), an unsatisfiable normalisation and a wrong exception pair in 6.2, an unsuitable choice of s and an untreated case $e_{45} = 0$ in 6.4, and a hypothesis of Remark 8(i) that fails where Lemma 9 uses it; in Section 5, Case 3, eq. (9) alone does not imply claim (12) (28 failures in 12544 model configurations), while (12) itself holds. None of these affects the theorem: every statement used by the argument is true, and each repair is local.

7. Scope and credits

Status of the main theorem. The theorem `eight_runners_unconditional_real` depends only on `propext`, `Classical.choice` and `Quot.sound`. This was re-observed on 2026-11-01 by a raw `lean` assembly (50 s, memory footprint 1.08 GB) together with a negative control; the hashes of the regenerated certificate sources match the records; the 27 kernel proofs were not rebuilt. The module is compiled outside `lake build`, so the label VERIFIED is used here for that raw compilation plus `#print axioms`, and not for a clean `lake build` of the repository. No whole-lake rebuild was run for this release, and the Lean sources are provided as is: the tree in `lean/` was not built as a whole, and the generated modules `E8Stage.P`p are not in it (they are regenerated by `code/eight`). The archived `lean/` tree does not build as a whole: `EightRunnersReal` imports generated modules (`E8Stage.P`p`.Asm`) that are not archived. The raw outputs of the re-observation are in `data/eight/recheck_2026_11_01`. We are not aware of another formal proof of the lonely runner conjecture for eight runners; this was not checked against the repository `formal-conjectures` or its discussion threads. A formalization of the case of seven runners by Chen [4] is described below.

This is a formalization of known theorems. The mathematical content of Theorem 1.1 is due to Rosenfeld [11]; that of Theorem 1.2 to Barajas and Serra [2]; the rounding step of Section 4 to Giri and Kravitz [7] (see also [1]). Rosenfeld's height bound is that of Malikiosis, Santos and Schymura [9]; the formalization avoids it, but the threshold 28^6 and the overall architecture are Rosenfeld's. Honglu Chen's repository [4] contains an earlier, independent Lean formalization of LRC(6) following Barajas and Serra; its statement was imported as an axiom, with a proved bridge to our formulation, as an interim input (`Chen7RunnerImport`). Theorem 1.1 does not depend on it. We did not re-check Chen's development, which uses a different Lean and Mathlib version.

The contributions of this paper are the Lean proofs, the certificate checker with its soundness proof, the reproduction procedure for the 27 certificates, the finite-model routes for $n = 4$ and for Section 5 of [2], and the list of errata. The formalization uses Lean 4.30.0 and Mathlib [15] pinned in the repository's `lake-manifest.json`. For background on the conjecture see [6, 14, 8, 10].

Provenance. The results were obtained with the assistance of an AI system (Claude, Anthropic). The questions and research directions came from the author. The Lean sources, the generator, the checkers and the text were produced by the AI system; the Lean theorems are checked by the Lean kernel, and the checks of 2026-11-01 were made by a further session of the same system. The author is responsible for all claims.

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